How Much Targeted Investment Does It Take to Top a Multi-Task Leaderboard? Minimum-Cost Manipulation under Aggregation Rules

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Paper under double-blind review

Abstract

A multi-task leaderboard ranks models by aggregating per-task scores, and developers can try to climb it by training on data related to some of the sub-datasets. We ask how much such targeted investment a developer needs to make one target model rank first, and how this depends on the aggregation rule. We model investment as raising the target's scores on chosen tasks within a cap, with all other models fixed, and measure it by the number of tasks touched (K), the total points raised (L_1) and the largest single-task jump (L_∞), always under a named cap scenario and together with the rate at which first place is attainable. For score rules we derive closed forms (mean, minimum, odd-n median) and solve power means and rank rules (Borda, plurality, Condorcet winner) exactly by mixed-integer programming. On public data (HELM MMLU v1.13.0, 79 models $\times$ 57 subjects; Open LLM Leaderboard, 4496 models $\times$ 6 tasks) the answer is conditional: for the ten closest challengers on HELM, the median-rule, minimum-rule and plurality costs are only 3–6% of the mean-rule cost in total points (paired medians), Borda and the Condorcet requirement are 41% and 17%, and geometric and harmonic means are 91% and 84%; the ordering of rules by cost is nearly unchanged across six cap scenarios, whereas first-place feasibility is not (the minimum rule is attainable for only 6 and 7 of 10 targets under the two tightest gain caps). Under evaluation noise the cheap rules have the largest relative spread of cost, though in points their spread is small. Contrary to our expectation, order-statistic rules force little breadth for near-top targets: with paired targets the median rule's K grows with the number of tasks (log–log slope of the mean cost 0.54, interval 0.51–0.56, against 0.84 for the mean rule) but stays far below the worst case of a majority of tasks, and the minimum rule's barely (0.16). On the six-task Open LLM Leaderboard the score scale matters, in paired comparisons over targets: the median rule's cost relative to the mean rule rises from 0.13 to 0.84 (ranks 11–30; paired difference 0.70, interval 0.61–0.82) and from 0.34 to 0.78 (ranks 31–100) when normalised scores are replaced by raw ones, but not beyond rank 100 (+0.008), and the minimum rule is dearer on raw scores in every rank group. We state where the conclusions fail and what we did not complete.

1 Introduction

Multi-task benchmarks are the standard way to compare machine-learning models: a leaderboard is a table of scores of many models on many sub-datasets, summarised by an aggregation rule, most often the (weighted) mean. Ranks influence adoption, publication and funding, so developers have a reason to act strategically. One such action is *targeted investment*: training the favoured model on data related to some of the sub-datasets so that its scores there rise. Public discussion of leaderboards and contamination (Singh et al., 2025; Zhou et al., 2023; Yang et al., 2023) documents the incentive, but not how much effort is needed to reach the top, nor how that depends on the way scores are aggregated.

We study exactly that. *How much targeted investment does it take for a given model to become first, and how does this vary with the aggregation rule?* We take the question to be *conditional by design*: the answer depends on a modelling assumption (how far a score can be raised, our cap scenarios),

on the rule, and on the target's starting position. We therefore report every result under a named cap, with three cost measures that are never merged, and with the rate at which first place is attainable at all.

Contributions. (1) A formal minimum-investment problem on a score matrix, with explicit cap scenarios, cost measures, margins and a set of rules (Section 3); the link to bribery in computational social choice is made explicit in Section 2. (2) Exact solvers for score rules and, by mixed-integer programming, for Borda, plurality and the Condorcet-winner requirement, tested against brute force, together with closed forms for the mean, the minimum and the odd-n median that we verify on every target of two public leaderboards. (3) An empirical study on HELM MMLU v1.13.0 (79 models $\times$ 57 subjects) and the Open LLM Leaderboard (4496 models $\times$ 6 tasks), with bootstrap intervals over targets and robustness analyses.

Findings (each conditional on the stated cap and data). (i) On HELM, for the ten challengers ranked 2–11 under the mean, reaching first place costs the mean rule a median of 162 points in total over 4 subjects under cap *ceil*; the median rule, the minimum rule and plurality need only 3–6% of that in total points (Section 4). (ii) This ordering of rules by cost hardly changes across six cap scenarios, but *feasibility* does: tight caps make first place unattainable for some targets under the minimum rule and the power means. (iii) The breadth that order-statistic rules force in theory (a majority of tasks for a target below the rival median everywhere) is only partly realised for targets near the top: with the same target for every rule, K under the median rule grows with the number of tasks n (log–log slope 0.54, 0.51–0.56; median $K = 6$ of 48 subjects), but far below $\lfloor n/2 \rfloor + 1$ and below the mean (0.84; $K = 13$ once per-task headroom is capped), and K under the minimum is nearly flat (0.16). (iv) The cheap rules have the largest relative spread of cost under evaluation noise. (v) On the six-task Open LLM Leaderboard the score scale matters for score rules in paired comparisons over targets: the median rule's ratio to the mean rule rises from 0.13 to 0.84 (ranks 11–30) and from 0.34 to 0.78 (ranks 31–100) under raw instead of normalised scores, but not beyond rank 100, and the minimum is dearer under raw scores in every rank group; rank rules are invariant. We also report what we did not complete (Section 5).

2 Related work

Aggregation and evaluation. Multi-task benchmarks such as MMLU (Hendrycks et al., 2020), BIG-Bench Hard (Suzgun et al., 2022) and the HELM framework (Liang et al., 2022) report a per-task score matrix that is then aggregated, usually by a (weighted) mean. Dehghani et al. (2021) show that task selection and aggregation choices can change the relative ordering of methods ("benchmark lottery"), and Lanctot et al. (2023) propose voting-theoretic aggregation for multi-task evaluation of agents and language models. These works treat the rule as the evaluator's choice; we instead ask what a *developer* must spend to win under each rule.

Strategic behaviour on leaderboards and contamination. Singh et al. (2025) analyse private testing and sampling asymmetries on Chatbot Arena; Zhou et al. (2023) and Yang et al. (2023) study training on benchmark data and the failure of decontamination, the real-world counterpart of "targeted investment". Chen et al. (2026) model benchmarking as a Stackelberg game between a designer and several competing developers who allocate post-training resources and characterise equilibria. We study a different, complementary object: the minimum cost for *one* target to be first, rule by rule, measured empirically on public leaderboards, with no equilibrium reasoning.

Computational social choice. Making a designated candidate win by changing the input at minimum cost is the bribery problem (Faliszewski et al., 2009a;b; Faliszewski & Rothe, 2016; Brandt et al., 2016), and shift bribery and margin of victory are close relatives. Here tasks play the role of voters, but the manipulated object is the candidate's own *cardinal* score vector (not voters' preferences), the number of voters n is small, costs are continuous and capped, and the question is empirical on real score matrices.

3 METHOD: INVESTMENT MODEL AND EXACT SOLVERS

Setting. A leaderboard is a score matrix $S \in \mathbb{R}^{M \times n}$ (M models, n sub-datasets or "tasks", higher is better) and an aggregation rule f that turns the scores of all models into a ranking. We fix a *target* model i and ask what the developer of i must invest to make it strictly first under f. The other $M-1$ rows are fixed: rivals do not react, so the question is one-sided and not an equilibrium analysis.

Investment model. The developer may *raise* the target's score on task t to any $y_t \in [x_t, c_t]$, where $x = S_{i\cdot}$ and $c_t \geq x_t$ is a cap (a stand-in for "training on data related to sub-dataset t"; scores are never lowered). A vector y is *successful* under f if the target is strictly first with margin ε: for score rules, $f(y) \geq \max_{j \neq i} f(S_{j\cdot}) + \varepsilon$ with $\varepsilon = 10^{-3}$ points (a tie-breaking margin); the main experiments also report a statistical margin $k \cdot \mathrm{SE}$ ($k = 1, 2$) from binomial sampling of the per-task item counts, so that "first" is not an artefact of evaluation noise. We report three costs of y, never pooled into each other: $K = |\{t : y_t > x_t\}|$ (tasks touched; scale free), $L_1 = \sum_t (y_t - x_t)$ (total points; for rule comparison also the ratio to the same target's cost under the mean rule), and $L_\infty = \max_t (y_t - x_t)$ (largest single-task jump, equal to the cost of a uniform improvement). If no y within the caps succeeds, the outcome is *infeasible*; it is reported as a rate and never replaced by a large number. The *minimum investment* is the minimum of a cost over successful y.

Cap scenarios (the main modelling assumption). The mapping from training effort to score gain is unknown, so the cap is a scenario and results are never pooled across scenarios: `ceil` ($c_t = 100$; isolates the rule), `front0`/`front2` (best existing score on t plus 0/2 points: "cannot beat the state of the art by much"), `gain5/10/20` ($c_t = x_t + h$), and `emp95`/`emp99` (per-task 95th/99th percentile of observed fine-tune-minus-base gains on the Open LLM Leaderboard; applied to HELM only as a labelled transfer assumption, so HELM uses cap-free and `gain-`h scenarios).

Aggregation rules. *Score rules:* arithmetic mean (the official rule of the Open LLM Leaderboard; equal-weight on HELM subjects), item-count-weighted mean (HELM's official "All Subjects"), power means with exponent $p < 1$ (geometric $p \to 0$, harmonic $p = -1$, interpolating to the minimum as $p \to -\infty$), median (even n: mean of the two middle values), trimmed mean, minimum. *Rank rules:* Borda (mean rank), plurality (first-place count, ties split), and the Condorcet-winner requirement. The last is a *labelled proxy* for pairwise rules: a Condorcet winner is first under every Condorcet-consistent rule (Copeland, Schulze, Kemeny), so its cost upper-bounds, but is not, their cost; we report how often the original leaderboard has no Condorcet winner.

Exact solvers. For each rule and cost we compute the exact minimum investment. Write $\tau = \max_{j \neq i} f(S_{j\cdot})$ and $b = \tau + \varepsilon$.

Proposition 1 (mean, minimum, median)**.** *(i)* Mean. *Let* $\mathrm{need} = nb - \sum_t x_t$. *If* $\mathrm{need} \leq 0$ *the cost is* 0*; otherwise first place is feasible iff* $\sum_t (c_t - x_t) \geq \mathrm{need}$*, the minimum* L_1 *is exactly* $\mathrm{need} = n \cdot (\mathit{gap}) + n\varepsilon$*, and the minimum* K *is the least* k *such that the* k *largest headrooms* $c_t - x_t$ *sum to at least* need *(so* $K = 1$ *when one task has enough headroom). (ii)* Minimum. *First place is feasible iff* $c_t \geq b$ *for all* t*; then* $L_1 = \sum_t \max(0, b - x_t)$*,* $K = |\{t : x_t < b\}|$ *and* $L_\infty = b - \min_t x_t$*. (iii)* Median, n odd. *Let* $m = (n+1)/2$*. The median of* y *is* $\geq b$ *iff at least* m *coordinates are* $\geq b$*; hence* $K = \max(0, m - |\{t : x_t \geq b\}|)$ *and* L_1 *is the sum of the* m *smallest gaps* $\max(0, b - x_t)$ *among tasks with* $c_t \geq b$ *(infeasible if fewer than* m *such tasks).*

Proofs are one line each: the mean constraint is $\sum_t y_t \geq nb$ and $L_1 = \sum_t y_t - \sum_t x_t$; $\min_t y_t \geq b$ iff every $y_t \geq b$; the median is $\geq b$ iff at least m order statistics are. Consequently the mean rule rewards *concentration* (one large jump on the task with most headroom), whereas the median and the minimum force *breadth*: a target below b on every task needs $K \geq \lceil n/2 \rceil$ under the median (for even n the larger of the two middle values must reach b, so $K \geq n/2$; $K = n/2 + 1$ is not always needed) and $K = n$ under the minimum. For power means with $p < 1$ the cost is minimised by equalising $y_t + s$ at a water level, found by bisection, and K by a greedy rule on $g(c_t) - g(x_t)$. Rank rules have no closed form: Borda, plurality and the Condorcet requirement are solved exactly as mixed-integer programmes over candidate "levels" at which a raised score passes a rival on a task (rival constraints generated lazily; HiGHS through SciPy). Every solver is unit-tested against brute force on small random instances, and Proposition 1 is checked numerically against the solvers on every target of both datasets (Section 4).

Table 1: Minimum investment for first place on HELM MMLU v1.13.0, ten challengers ranked 2–11 under the mean rule: median [95% bootstrap CI over targets] of tasks touched K, total points L_1 and largest single-task jump L_∞, over targets for which first place is attainable; "feasible" counts attainable targets out of 10 (infeasible targets are counted, not imputed). Caps: ceil (100) and gain+10; the other caps are in the appendix table. Margin $\varepsilon = 10^{-3}$. Source: `results/helm_main.csv`; script `tab_rule_summary.py`.

rule	feasible	K	L_1	L_∞
cap: ceil (100)				
mean	10/10	4 [1, 6]	162 [42, 222]	2.87 [0.73, 3.95]
item-count-weighted mean	10/10	1 [1, 2]	24.9 [11.3, 37.7]	2.74 [1.24, 3.68]
geometric mean	10/10	4 [2, 6]	147 [38, 202]	3.21 [0.89, 4.11]
harmonic mean	10/10	4 [2, 6]	138 [38, 185]	3.71 [1.15, 4.34]
median	10/10	4.5 [2, 9]	4.73 [0.38, 16.64]	1.49 [0.38, 2.78]
min	10/10	2 [1, 3]	5.29 [2.22, 23.86]	3.83 [1.64, 12.07]
Borda	10/10	17.5 [10, 27]	54.6 [21.1, 100.5]	2.02 [1.21, 3.07]
Condorcet winner	10/10	17 [8, 22]	22.6 [4.6, 55.4]	1.74 [0.98, 3.03]
plurality	10/10	7 [4, 8]	5.18 [1.42, 11.42]	1.03 [0.50, 2.26]
cap: gain+10				
mean	10/10	16.5 [5, 23]	162 [42, 222]	2.87 [0.73, 3.95]
item-count-weighted mean	10/10	5 [2, 9]	42.7 [12.2, 79.1]	2.74 [1.24, 3.68]
geometric mean	10/10	16 [4, 21]	155 [38, 209]	3.21 [0.89, 4.11]
harmonic mean	10/10	15.5 [4, 20.0125]	153 [38, 197]	3.71 [1.15, 4.34]
median	10/10	4.5 [2, 9]	4.73 [0.38, 16.64]	1.49 [0.38, 2.78]
min	7/10	2 [1, 3]	2.41 [2.02, 7.57]	2.41 [1.21, 4.64]
Borda	10/10	19.5 [10, 30]	54.6 [21.1, 100.5]	2.02 [1.21, 3.07]
Condorcet winner	10/10	17 [8, 22]	22.6 [4.6, 55.4]	1.74 [0.98, 3.03]
plurality	10/10	7 [4, 8]	5.18 [1.42, 11.42]	1.03 [0.50, 2.26]

Targets and uncertainty. Targets are fixed ex ante: the challengers ranked 2–11 under the mean rule (primary), plus strata by initial rank band and gap. Uncertainty comes from bootstrap over targets, over tasks, and Monte-Carlo resampling of evaluation noise, as each claim demands; every aggregate is reported with its interval, its cap scenario and its feasibility rate.

4 Experiments

Data and protocol. **HELM MMLU** v1.13.0 (release summary dated 2025-01-10): 79 models $\times$ 57 subjects, exact-match accuracy in points, no missing cells; the item count per subject ranges from 100 to 1534. **Open LLM Leaderboard** (`contents`, commit `9c09a7c...`, 2025-03-20): 4576 rows reduced to 4497 models after removing duplicate names (best variant kept) and to 4496 after removing the one flagged model; $n = 6$ normalised task scores (the leaderboard's own normalisation, which removes random-guess baselines before the official arithmetic mean). Models on both boards are not independent (merges, same-base fine-tunes); we say where this matters. The primary targets are fixed in advance as the ten challengers ranked 2–11 under the mean rule ("top-10"); results by initial rank use all targets, and every interval is a 95% percentile bootstrap over targets unless stated. Cap scenarios and costs are those of Section 3; *gain*$+h$ allows at most h points per subject, *front*$+c$ the best existing score on the subject plus c. The margin is $\varepsilon = 10^{-3}$ points unless a statistical margin is stated.

The closed forms are exact. Proposition 1 agrees with the exact solvers on every target checked: for the mean, the minimum and the odd-n median, the maximum absolute error in L_1 and K is 0 over 78–79 HELM and 4495 OLLB targets (and $\leq 10^{-7}$ for L_∞, the bisection tolerance); the cost of every checked target never decreases when the cap is tightened (0 violations in 1896 HELM and 3600 OLLB checks; Table 2, appendix).

Rule $\times$ cap: cost differs strongly, feasibility depends on the cap. Table 1 and Figure 1 give the headline on HELM. Under cap *ceil*, a mean-rule win costs a median of 162 points in total (CI

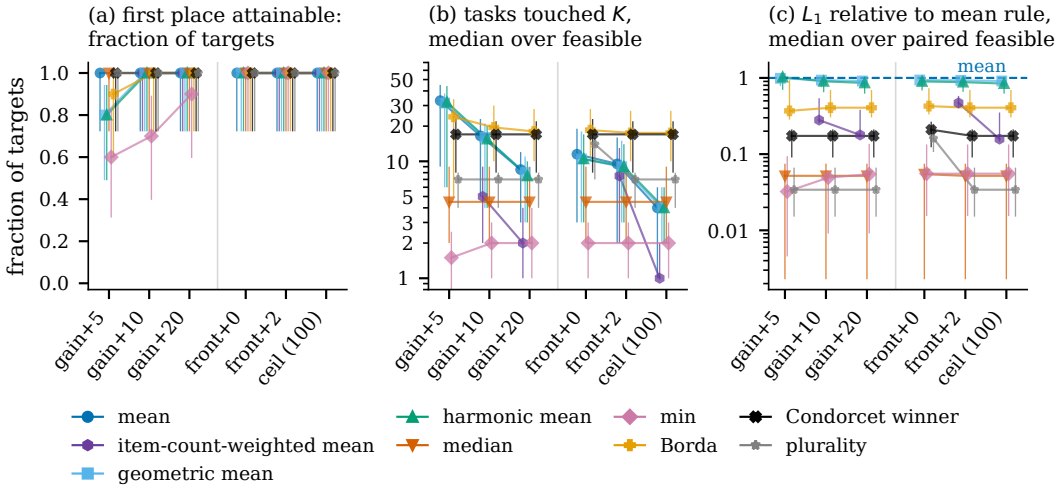


Figure 1: Cost of first place per aggregation rule and cap scenario, HELM MMLU v1.13.0 (79 models $\times$ 57 subjects), targets = the ten challengers ranked 2–11 under the mean rule (the mean rule's own value is the dashed reference in (c)). Caps: *gain*+h (a score may rise by at most h points per subject), *front*+c (up to the best existing score on the subject $+c$), *ceil* (100); they are separate scenarios, not one scale, and lines connect only caps of the same family. (a) Fraction of the 10 targets for which first place is attainable at all (Wilson 95% interval). (b) Median number of tasks touched K over feasible targets; (c) median per-target ratio $L_1^{\text{rule}}/L_1^{\text{mean}}$ over targets feasible under both rules (ratio < 1: cheaper than the mean). Medians are over feasible targets only, so points with different feasible fractions in (a) are not comparable in (b,c). Bars in (b,c): 95% percentile bootstrap over targets (2000 resamples); with 10 targets they are wide. The item-count-weighted mean (HELM's official aggregate) was run under ceil, front+2, gain+10 and gain+20 only. Margin $\varepsilon = 10^{-3}$. Source: `results/helm_main.csv`, `helm_wmean.csv`; script `fig_rule_cap.py`.

42–222; $K = 4$) for the ten closest challengers, i.e. $n \cdot (\text{gap} + \varepsilon)$ by Proposition 1, which is why L_1 under the mean scales with the number of tasks. The median rule needs 4.7 points ($K = 4.5$), the minimum rule 5.3 ($K = 2$), plurality 5.2 ($K = 7$), the Condorcet requirement 22.6 ($K = 17$) and Borda 54.6 ($K = 17.5$). The paired per-target ratio to the mean rule (Fig. 1c) has median 0.052 for the median rule, 0.055 for the minimum, 0.034 for plurality, 0.17 for the Condorcet requirement, 0.41 for Borda, and 0.91 and 0.84 for the geometric and harmonic means (CIs in the figure; with ten targets they are wide, e.g. 0.002–0.075 for the median). K and L_1 measure effort in score space, not training cost. Across the six caps this ordering of rules is nearly unchanged: the median ratios stay within narrow bands (median rule 0.052–0.055, minimum 0.032–0.055, Condorcet requirement 0.17–0.21, Borda 0.37–0.43), and the exception is plurality, which rises from 0.034 to 0.16 under *front*+0. What *does* depend on the cap is feasibility (Fig. 1a): under *gain*+5 first place is attainable for only 6/10 targets under the minimum rule, 8/10 under the geometric and harmonic means and 9/10 under Borda, against 10/10 under the mean (at *gain*+10: 7/10 for the minimum). The cost medians are over feasible targets only, so they must be read together with these rates.

Official HELM rule: item-count weighting. The official HELM "All Subjects" score weights subjects by their item count. For the same top-10 targets the paired L_1 ratio of this rule to the equal-weight mean is 0.16 (CI 0.14–0.33) under *ceil*, 0.28 under *gain*+10 and 0.47 under *front*+2 (one-subject investment suffices under *ceil*, $K = 1$): a target strong on heavy subjects needs far less under the official rule, and the advantage shrinks as caps tighten.

Cost against initial rank. Figure 2 shows cost against the target's initial rank under the mean rule (cap *ceil*). On HELM, L_1 increases steadily with rank for every rule, but rank rules saturate in breadth: plurality never needs more than 9 subjects and the Condorcet requirement never more than 29 (the majority of 57), while Borda needs up to 54 and the mean up to 46 (every target is feasible under *ceil*). For targets ranked 31 or lower the median rule has median $K = 26$, near the majority bound of Proposition 1; the maximum over HELM targets is $K = 29 = (n + 1)/2$. On the Open

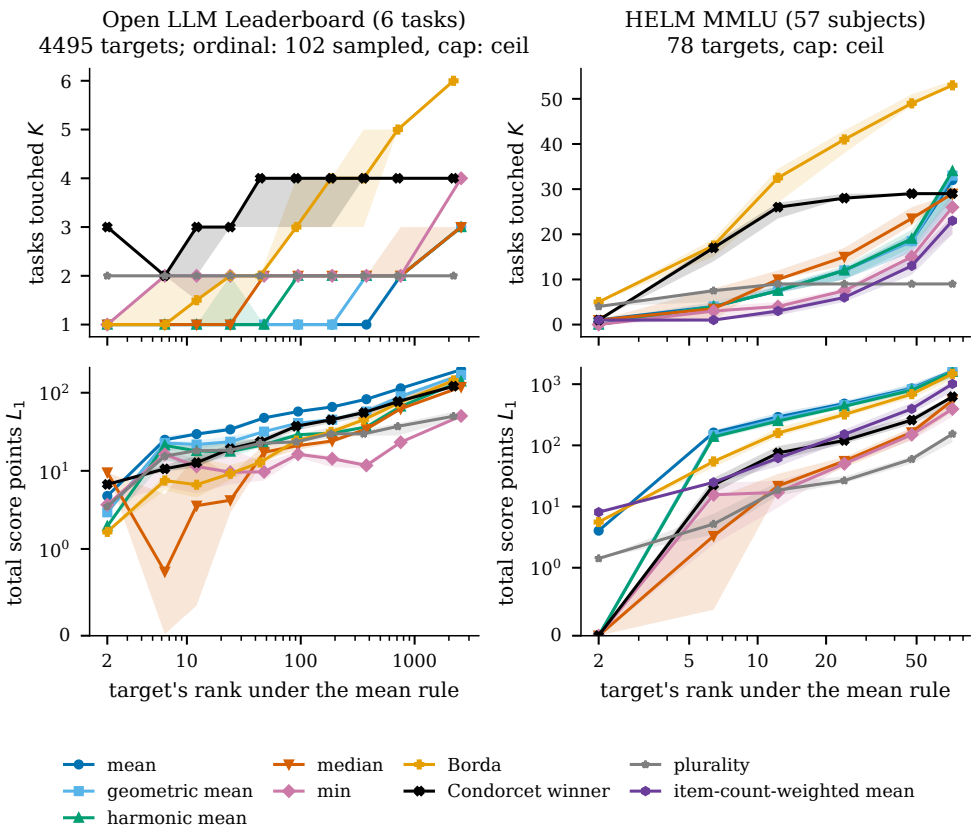


Figure 2: Minimum investment under cap *ceil* (scores up to 100) against the target's rank under the mean rule (log axis). Left: Open LLM Leaderboard (6 tasks; 4495 targets for the score rules; Borda, Condorcet and plurality are computed for 102 sampled targets: the top-ranked challengers and a log-rank-uniform sample); right: HELM MMLU (57 subjects; 78 targets, score and rank rules and the item-count-weighted mean). Lines: median in rank bins $\{2\}, \{3\text{–}4\}, \{5\text{–}8\}, \ldots$ plotted at the geometric mean rank of the bin; bands: 95% percentile bootstrap over the targets of the bin (1000 resamples; bins with fewer than 3 targets omitted, except rank 2). Top: tasks touched K; bottom: total score points L_1 (symlog axis, linear below 1). A value of 0 means the target is already first under that rule. One OLLB Condorcet target hit the time limit and is excluded; no other target is infeasible under *ceil*. Source: `results/ollb_closed.csv`, `ollb_ordinal_ceil.csv`, `helm_main.csv`, `helm_wmean.csv`; script `fig_rank_rules.py`.

LLM Leaderboard ($n = 6$, score rules only) a handful of tasks already suffices at every rank: $K \leq 6$ for all rules and targets, medians 2–3 (for the ten closest challengers, mean: $K = 1$, $L_1 = 25$ points; median rule: $L_1 = 0.9$; Table 6).

Does breadth grow with the number of tasks? We drew $n \in \{3, 6, 12, 24, 48\}$ subjects from the 57 (200 draws each) and priced the model ranked 2 or 5 *under the mean rule* in that subset for every rule, so that all rules price the same models (Figure 3; *front*+2 in the appendix). For the rank-5 target under *gain*+10 the median K at $n = 3, 6, 12, 24, 48$ is $1, 2, 4, 7, 13$ for the mean (slope 0.84), $1, 1, 2, 3, 6$ for the median rule (slope 0.54), $1, 2, 2, 2, 2$ for the minimum (slope 0.16), $2, 2, 3, 4, 6$ for plurality (0.51) and $2, 2, 4, 8, 16$ for Borda (0.82); for the rank-2 target $K = 1$ under the mean, the median and the minimum at every n. The total points L_1 grow linearly in n for the mean (slope 1.00, 0.98–1.02) and the geometric mean (0.98), but hardly for the median (0.33), the minimum (0.06) and plurality (0.14). The minimum has no feasible win in 10–31% of the draws, and the medians are over feasible draws. Slopes are of the log mean cost over feasible draws on $\log n$ (95% bootstrap intervals over draws: 0.82–0.86 for the mean rule and 0.51–0.56 for the median rule in K); slopes of the integer medians are 0.92, 0.68 and 0.20. Two conclusions follow. First, the mean

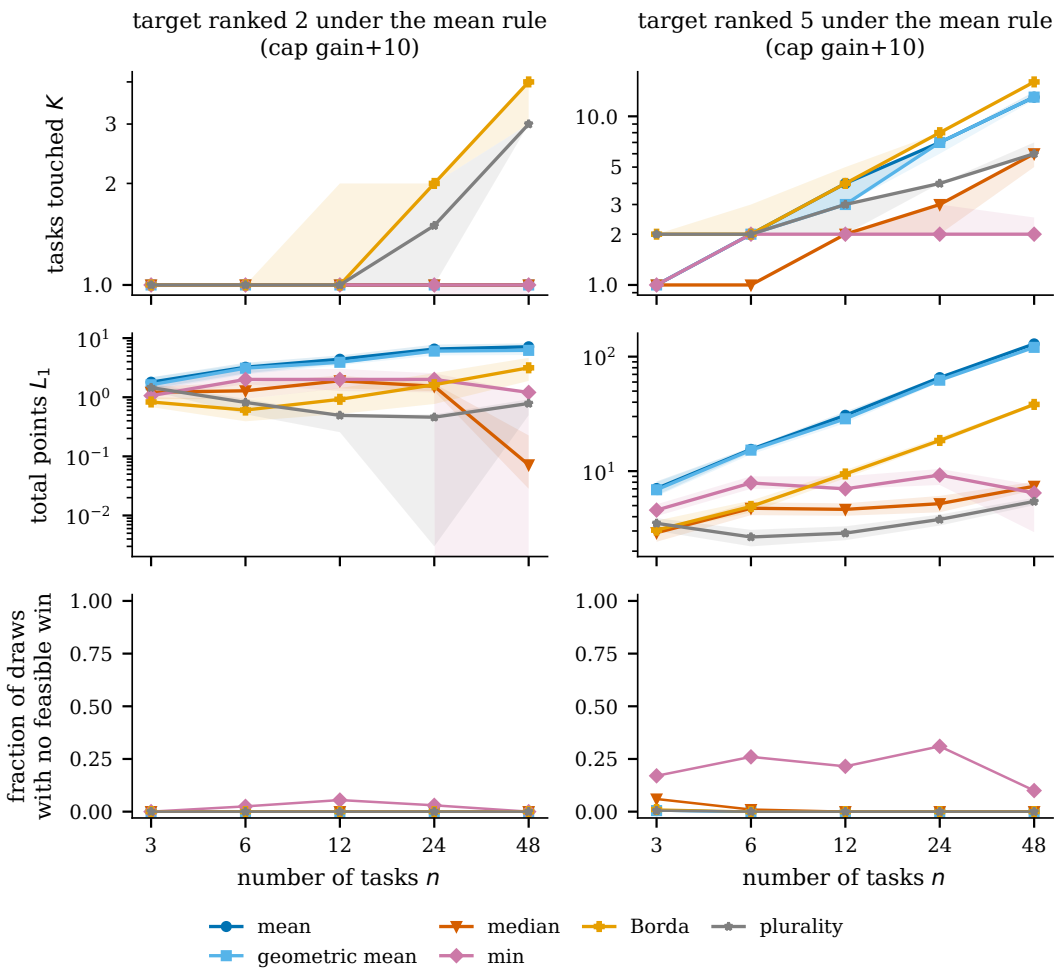


Figure 3: Investment against the number of tasks n, HELM MMLU, cap *gain+10*: n subjects are drawn at random from the 57 (200 draws per n). In every draw the targets are the models ranked 2 and 5 under the *mean* rule in that subset, so all rules price the same models. Lines: median over draws, with 95% percentile bootstrap bands over draws. Top: tasks touched K; middle: total points L_1 (draws in which the target already wins cost 0 and cannot be shown on the log axis); bottom: fraction of draws in which first place is infeasible under the cap (medians are drawn only where finite). Source: `results/summary_helm_tasksubpaired_gain10_by_n.csv`; script `fig_taskscale.py --paired`.

rule's K *does* grow with n once headroom per task is capped ($K \approx \text{need}/h$ with $\text{need} = n{\cdot}\text{gap}$), contrary to our prior expectation that K under the mean is independent of n; it is independent only when one task has the headroom (cap *ceil*). Second, the order-statistic rules do need more tasks as n grows (median rule: yes), but far fewer than the worst case $\lfloor n/2 \rfloor + 1$ (6 of 48, not 25) and never more than the mean; the minimum hardly forces breadth (slope 0.16). Our expectation that the median and the minimum scale linearly in K is therefore not borne out for near-top targets: scores are strongly correlated across subjects, so such targets already clear the rival median on many subjects; the median-rule K is $\approx (n+1)/2 - n_{\uparrow}$, with $n_{\uparrow}$ the number of subjects where the target already reaches the strongest rival's median (correlation 0.95 over non-leader targets).

Robustness. *Noise.* We ran a parametric bootstrap of evaluation noise (100 draws of every per-subject score using its binomial standard error) and re-solved (cap *gain*+10, top-10 targets); the median over targets of the coefficient of variation of L_1 across draws is 0.17 for the mean, 0.18 for the geometric mean, 0.34 for Borda and for the Condorcet requirement, 0.44 for plurality, 0.68

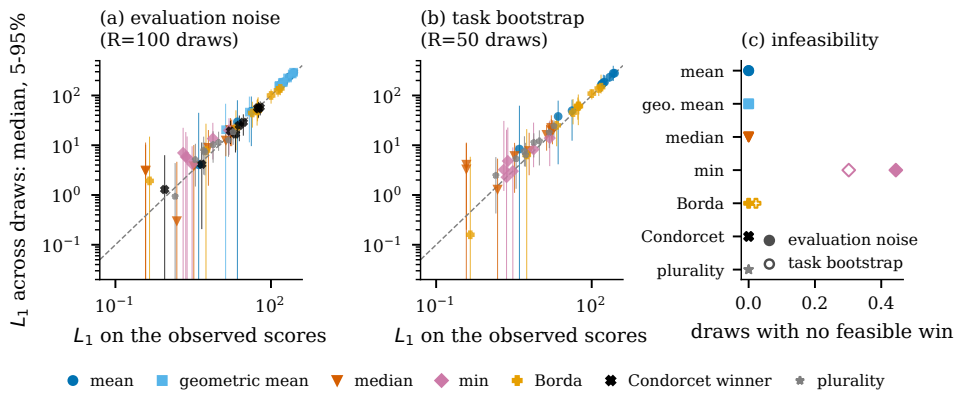


Figure 4: Stability of the minimum investment L_1 (HELM MMLU, cap gain+10, ten challengers ranked 2–11 under the mean rule) under (a) evaluation noise (100 draws of per-subject scores) and (b) a bootstrap over tasks (50 draws). Each marker is one (rule, target); abscissa: L_1 on the observed scores; ordinate: median over draws with the 5–95% range of the draws (finite draws only). Dashed: identity; targets already first on the observed scores ($L_1 = 0$) are not shown. (c) Fraction of draws without a feasible win, by rule (filled: noise; open: task bootstrap, run for five of the seven rules). Source: `results/summary_helm_{noise,taskboot}_gain10_{per_target,by_rule}.csv`; script `fig_robust.py`.

for the median and 0.75 for the minimum, and the minimum has no feasible win in 44% of draws (Fig. 4). The coefficient of variation is inflated for small costs; in points, the median 5–95% range of L_1 is 89 (mean), 97 (geometric mean), 56 (Borda), 24 (Condorcet), 15 (median and minimum) and 10 (plurality), against median costs of 174, 170, 50, 22, 6.5, 7.0 and 7.6: small, but about twice the cost for the median and minimum rules. *Task bootstrap* (50 draws over subjects) gives the same picture (minimum infeasible in 30% of draws, coefficient of variation 0.58 for the median). *Statistical margin.* Requiring the target to lead by $k = 1$ or 2 standard errors of the difference (0.6 points per k on HELM; approximate for the median and the minimum) leaves the paired ratios nearly unchanged at *ceil*: median rule 0.065 and 0.078 against 0.052 at $\varepsilon = 10^{-3}$, minimum 0.055 and 0.051, geometric mean 0.91 throughout. On the Open LLM Leaderboard (margin $k \times 0.96$ points) the median rule's ratio to the mean at *ceil* rises from 0.033 to 0.088 and 0.127, the minimum's falls from 0.61 to 0.44 and 0.41. The margin variants were run for score rules only; for rank rules a significance margin is not defined by ε, and their noise behaviour is the resampling experiment above.

Open LLM Leaderboard: caps, empirical envelope and score normalisation. Cap dependence on the six-task board is stronger than on HELM (Figures 6 and 7, appendix), because six tasks leave little room to spread a gain (top-10 targets, normalised scores; `summary_ollb_by_rule_group.csv`). Under *gain*+5 the share of targets for which first place is attainable is 1.0 for the mean and the median but 0.4 for the geometric and harmonic means, 0.3 for the minimum and 0.3 for the trimmed mean; under *gain*+10 it is 0.7 for the minimum and, on the sampled targets below, low for plurality. The ratio to the mean-rule cost at *ceil* is 0.03 for the median, 0.23 for Borda, 0.51 for the Condorcet requirement, 0.61 for the minimum, 0.64 and 0.76 for the harmonic and geometric means, 0.73 for plurality and 0.88 for the trimmed mean (under *emp95*, and at *ceil* on the top-500 slice); the ordering of the cheap rules is the same as on HELM, but only the median is much cheaper than the mean. *Empirical envelope.* The *emp95* cap is the per-task 95th percentile of the gains between 540 fine-tuned models and their bases on this leaderboard (applied to HELM it would be a transfer assumption, so we do not). Under *emp95*, first place is attainable for 10/10 top-10 targets under the mean, the median, the trimmed mean and Borda, 9/10 under the geometric mean and plurality, 5/10 under the Condorcet requirement and 4/10 under the harmonic mean and the minimum; under *emp99* the minimum is attainable for 5/10 and every other rule for 10/10. Even when the mean-rule win is attainable it needs 4.2 points per task, whether that is plausible depends on the envelope assumption, not on the data. *Raw versus normalised scores.*

Paired per target (cap *ceil*; raw minus normalised ratio $L_1^{\text{rule}}/L_1^{\text{mean}}$, 95% bootstrap intervals over targets; `summary_ollb_raw_vs_norm_paired.csv`), the median rule's ratio rises from 0.13 to 0.84 for ranks 11–30 ($+0.70$, 0.61–0.82) and from 0.34 to 0.78 for ranks 31–100 ($+0.45$, 0.43–0.46), but not for the 4396 targets beyond rank 100 ($+0.008$); for the ten closest challengers it rises from 0.03 to 1.10 ($+0.93$), with an interval (-0.15 to 1.27) that includes zero. The minimum is dearer under raw scores in every group ($+0.24$ beyond rank 100); geometric and harmonic means have mixed signs (Figure 10a, Table 7, appendix). The score scale thus changes which score rule is cheap for near-top models; rank rules are invariant to per-task affine rescaling. Slices (no merged models; top 500) leave the other ratios unchanged and move the minimum (0.49), Borda (0.19) and plurality (0.51).

Ordinal rules on the Open LLM Leaderboard. Exact mixed-integer solutions for Borda, plurality and the Condorcet requirement on 4496 models are expensive, so these rules were run on a sample of 102 targets (the 20 top challengers plus 100 targets uniform in log-rank) at *ceil*, *front*$+2$, *gain*$+10$ and *emp95*. On the sample, first place is unattainable under *gain*$+10$ for 37 targets under Borda, 62 under the Condorcet requirement and 89 under plurality, and under *emp95* for 43, 86 and 63; among the ten closest challengers plurality is attainable for only 3 under *gain*$+10$ and the Condorcet requirement for 5 under *emp95*. The sample favours low-ranked models. The original board has no Condorcet winner under the six-task majority relation (HELM has one), so the Condorcet requirement there is a proxy and an upper bound on Copeland, Schulze and Kemeny.

5 Discussion and limitations

What the evidence supports. In score space and under the stated caps, the aggregation rule changes the minimum targeted investment by more than an order of magnitude: the mean rule rewards one large jump summed over the leaderboard and is expensive in total points, while rules based on order statistics or on counting (median, minimum, plurality) are cheap in total points for near-top targets. The minimum and the power means are exposed to caps (first place becomes unattainable for some targets), and the median's cheapness comes with a large relative spread of cost under evaluation noise (the minimum's is larger still). The common rule of thumb that a median or minimum "forces breadth" holds for weak targets (worst case attained: $K = 29$ on HELM), but for challengers near the top the median rule's K grows only slowly with the number of tasks and the minimum rule's hardly at all. On the six-task Open LLM Leaderboard the cost of the median rule and the minimum relative to the mean depends on the score scale in paired comparisons (median rule: ranks 11–100; minimum: every group), not for the median rule beyond rank 100.

Where conclusions do not hold or were not tested. (1) *Modelling assumption.* Investment is a score increase within a cap; we make no statement about the real cost of fine-tuning, whether gains transfer, or contamination. empirical caps from fine-tune/base differences (540 pairs) exist only for the Open LLM Leaderboard, and whether a required gain is plausible depends on that envelope assumption, not on the data alone. (2) *One-sided.* Rivals do not react; this is not an equilibrium analysis like Chen et al. (2026). (3) *Small n.* With six tasks (OLLB) the median, minimum and plurality are coarse; rule-versus-n claims come from HELM subsets. (4) *Few targets, dependent models.* The primary target set has ten models, so intervals are wide, and models share bases and training data; item-level noise is addressed by resampling and margins, not by modelling dependence. (5) *Proxy for pairwise rules.* The Condorcet requirement upper-bounds Copeland, Schulze and Kemeny; their exact costs are not reported. (6) *Not completed.* Ordinal rules on the Open LLM Leaderboard ran on a 102-target sample and not at every cap; noise, bootstrap and task-subset robustness ran on HELM only; there is no statistical margin for the rank rules and no exact Copeland (Appendix B).

Reproducibility statement

All numbers, tables and figures are produced by the released code from two pinned public snapshots: HELM MMLU release v1.13.0 and the Open LLM Leaderboard `contents` dataset at commit `9c09a7cae43334062a82cb164f2ef255013dafa2` (parquet SHA-256 `f13a7061...dca5`). The reproduction notes accompanying the submission

(`code/REPRODUCE.md`, with `code/RESULTS.md` mapping every claim to its result file) give the environment and package versions, data checksums and licence note, seeds, commands and measured run times (about 12 CPU-hours in total, mostly the exact mixed-integer programmes), and map every table and figure to the script that generates it and the result file it reads. Solvers are deterministic and tested against brute force; random components use fixed seeds (targets, bootstrap and robustness drivers seed 0). The solvers' unit tests and the closed-form checks are part of the code. The code is released under the Apache License 2.0 (`code/LICENSE`).

ETHICS STATEMENT

The study uses only public aggregate benchmark scores; no model was trained or evaluated and no human subjects or personal data are involved. The analysis shows how cheaply some aggregation rules can be gamed in a stylised score-space model; we see it as input for benchmark designers and not as a recipe, since it makes no statement about real training cost and cannot identify any actual developer's behaviour.

AI USE STATEMENT

This study was carried out end to end by generative AI agents (large language models of the Claude family, operating as cooperating research roles with shell, code-execution and literature-search tools). The task description and the frozen input files were supplied to them; no human wrote text, code or analysis. We used generative AI tools for developing the conceptual framework, formulating the mathematical statements, designing the methodology and the experiments, implementing the methods, cleaning and reformatting the public leaderboard data, interpreting the results, drafting the paper, creating the figures and tables, writing and editing the code, searching for and summarising literature, and formatting references. No synthetic data stands in for measurements: every number, table and figure is computed by the released code from the public snapshots. All AI-assisted work was reviewed by other AI roles of the same pipeline, not by humans. We take responsibility for the final content of this work.

FORBIDDEN-MATERIAL STATEMENT

The task names a published paper on this question that must not be searched for or read. No search for it or for pages discussing it was made while writing this paper, and none was encountered; the literature consulted is the starter list and classical computational social choice work on bribery.

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A Additional results

Table 2: Closed-form propositions against the exact solvers: maximum absolute error over all checked targets (number of checks in parentheses; for the monotonicity row, the number of violations out of the checks). τ: best rival score under the rule; $b = \tau + \varepsilon$; x_t: the target's scores. P3 is stated for odd n under cap ceil and is therefore not available for the six OLLB tasks; the uncapped L_∞ check of P1 only applies to the few targets that need no cap. Source: `results/theory_checks.csv`; script `tab_theory.py`.

	statement	HELM: max. err. (checks)	OLLB: max. err. (checks)
P1 mean	$L_1 = n(\tau+\varepsilon) - \sum_t x_t$	0.0e+00 (78)	0.0e+00 (4495)
P1 mean	L_∞ (uncapped) $= \tau + \varepsilon - \bar{x}$	7.6e-08 (3)	9.3e-08 (4346)
P1 mean	$K \geq \lceil L_1 / \text{max headroom} \rceil$ (violations)	0.0e+00 (78)	0.0e+00 (4495)
P2 min	$L_1 = \sum_t \max(0, b - x_t)$	0.0e+00 (78)	0.0e+00 (4495)
P2 min	$K = \#\{x_t < b\}$	0.0e+00 (78)	0.0e+00 (4495)
P2 min	$L_\infty = b - \min_t x_t$	9.2e-08 (78)	9.1e-08 (4495)
P3 median	$L_1 =$ sum of the $(n+1)/2$ smallest gaps	0.0e+00 (79)	n/a
P3 median	$K = \max(0, (n+1)/2 - \#\{x_t \geq b\})$	0.0e+00 (79)	n/a
P4 monotone	cost never decreases when the cap tightens (violations; mean, geo. mean, median, min)	0 of 1896	0 of 3600

B What was not completed

Ordinal rules on the Open LLM Leaderboard were run on a sample of 102 targets, not all 4496 targets, and not at *gain*$+5$, *gain*$+20$ and *front*$+0$, and at *emp99* only for the ten closest challengers; the trimmed mean was not run on the raw and non-merged variants; noise, task-bootstrap and task-subset robustness were run only for *gain*$+10$ (noise, bootstrap) and *gain*$+10$ and *front*$+2$ (task subsets) on HELM, not on the Open LLM Leaderboard; there is no statistical-margin variant for the rank rules and no exact Copeland; the Open LLM Leaderboard `results` sub-tasks were not analysed. A solver time limit left one Condorcet cost undefined at *ceil* (and one K at *front*$+2$). Where a table lists only some rules or caps, the others were not run.

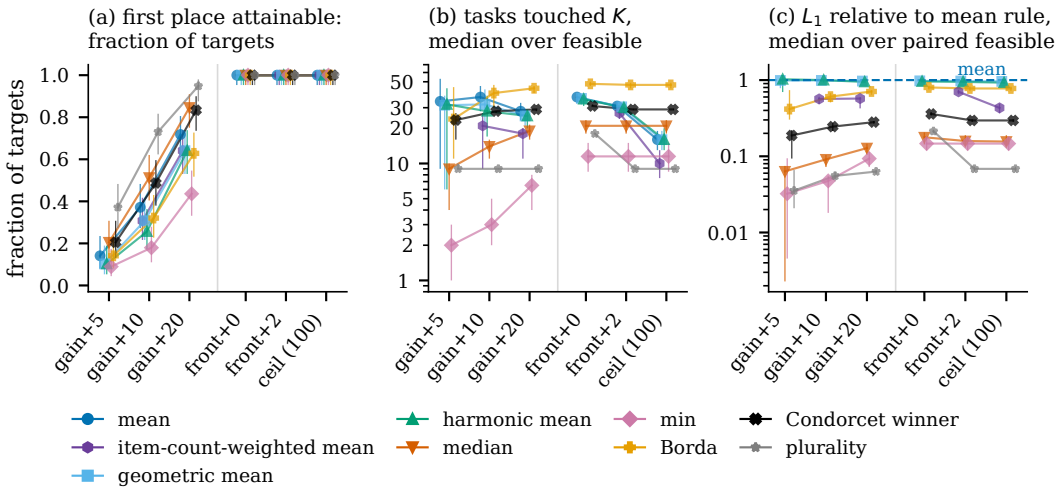


Figure 5: As the headline rule–cap figure, but for all 78 challengers (every model except the mean-rule leader) of HELM MMLU v1.13.0; same encoding, intervals and exclusions (Wilson 95% in (a), percentile bootstrap over targets in (b,c)). Source: `results/helm_main.csv`, `helm_wmean.csv`; script `fig_rule_cap.py --targets all`.

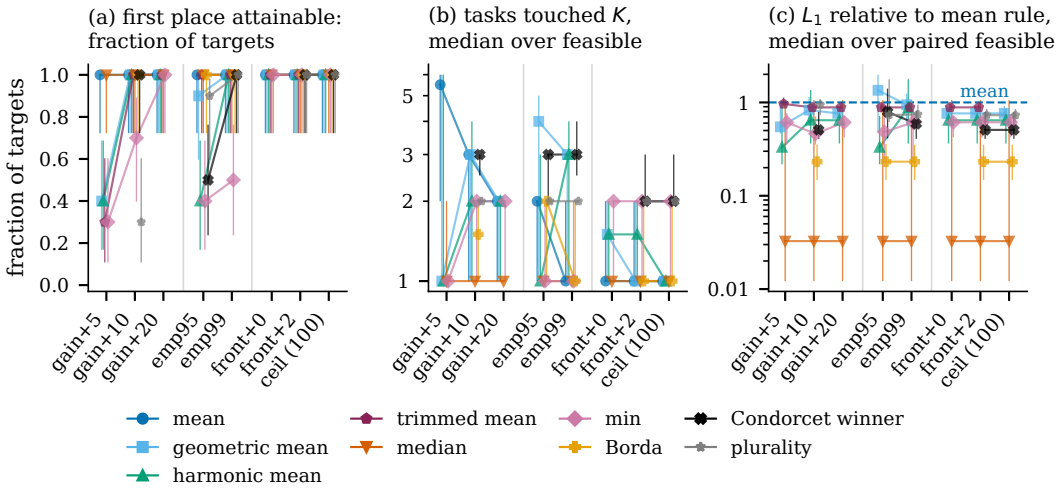


Figure 6: As the HELM rule–cap figure, for the Open LLM Leaderboard (6 tasks, 4496 models): the ten challengers ranked 2–11 under the mean rule. Caps: gain+h, *emp95*/*emp99* (per task, the 95th/99th percentile of the observed fine-tune-minus-base score gain, 540 pairs), front+c, ceil. Score rules were run under all caps; the rank rules (Borda, Condorcet winner, plurality) only under ceil, front+2, gain+10 and emp95, and emp99 for the top 20 (sampled targets: 102 models, including the ten shown). (a) Wilson 95% interval; (b,c) medians over feasible targets with 95% bootstrap over the 10 targets; infeasible targets are counted in (a), not imputed. With $n = 6$ tasks $K \leq 6$ and the intervals are very wide. Source: `results/ollb_closed.csv`, `ollb_score_caps.csv`, `ollb_score_emp.csv`, `ollb_ordinal_*.csv`; script `fig_rule_cap.py --dataset ollb`.

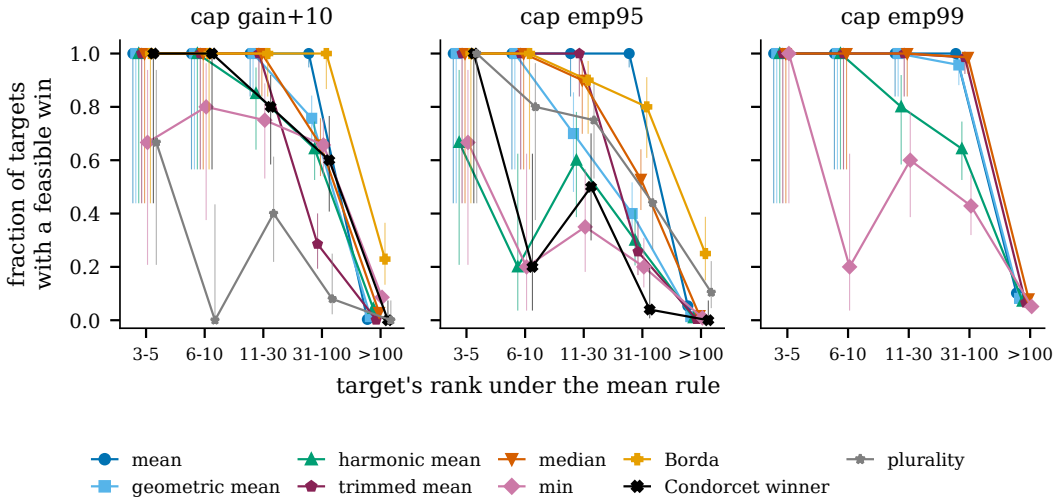


Figure 7: Open LLM Leaderboard: fraction of targets for which first place is attainable inside the cap, by the target's initial rank band under the mean rule (3–5, 6–10, 11–30, 31–100, >100; rank 2 is one model and omitted), for caps gain+10, *emp95* and *emp99* (per-task 95th/99th percentile of observed fine-tune gains). Bars: Wilson 95% intervals. Score rules are evaluated on every model of the band (3, 5, 20, 70, 4396 targets); Borda, Condorcet and plurality on a sample (3, 5, 20, 25, 48 targets) and, in this summary, not under emp99 (that run covers only the top 20), so their curves are noisy. Whether a feasible investment is plausible depends on the envelope assumption, not on the data alone. Source: `results/summary_ollb_by_rule_group.csv`; script `fig_feasible_ollb.py`.

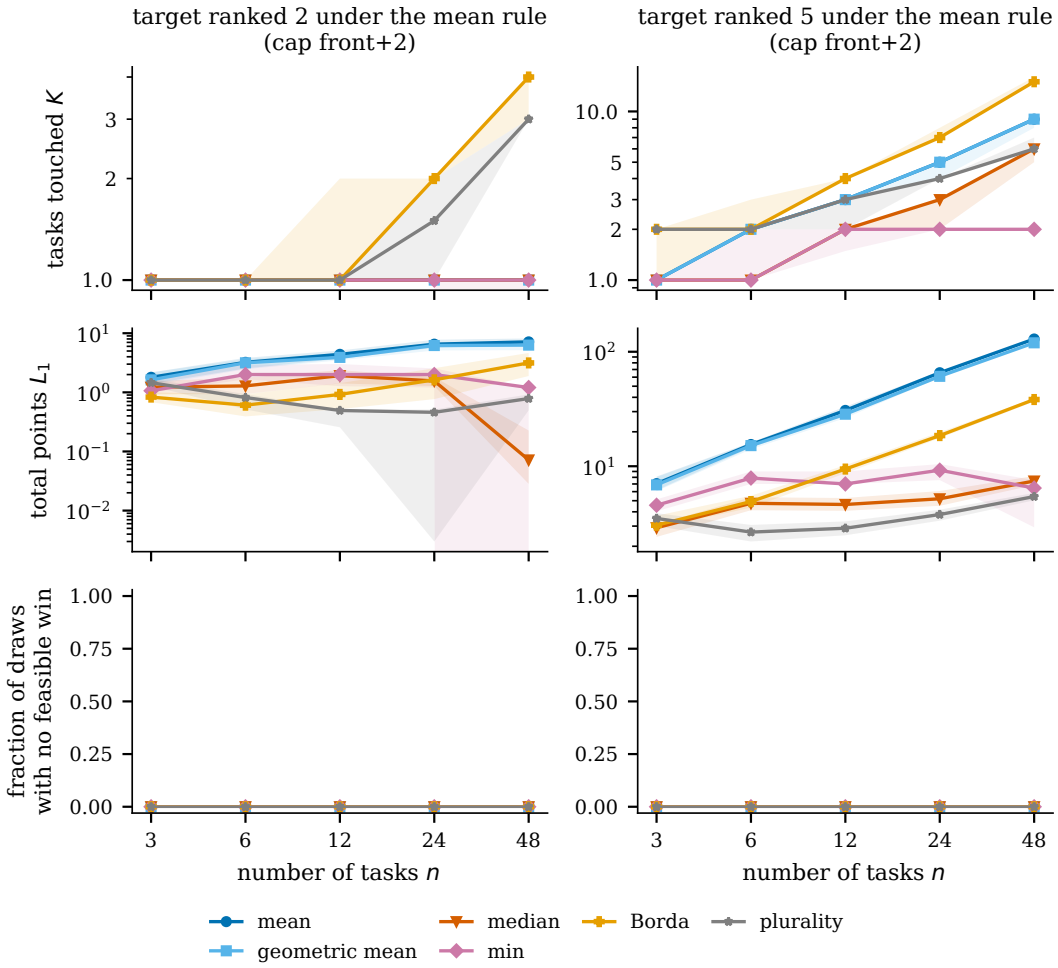


Figure 8: Investment against the number of tasks n, HELM MMLU, cap *front+2*: n subjects are drawn at random from the 57 (200 draws per n). In every draw the targets are the models ranked 2 and 5 under the *mean* rule in that subset, so all rules price the same models. Lines: median over draws, with 95% percentile bootstrap bands over draws. Top: tasks touched K; middle: total points L_1 (draws in which the target already wins cost 0 and cannot be shown on the log axis); bottom: fraction of draws in which first place is infeasible under the cap (medians are drawn only where finite). Source: `results/summary_helm_tasksubpaired_front2_by_n.csv`; script `fig_taskscale.py --paired`.

Table 3: As the main HELM table (ten challengers ranked 2–11 under the mean rule), for the caps front+2, gain+20 and the two caps of the main table (appendix). Cells: median [95% bootstrap CI over targets] of K, L_1, L_∞ over feasible targets; infeasible targets are counted, not imputed. Source: `results/helm_main.csv`; script `tab_rule_summary.py`.

rule	feasible	K	L_1	L_∞
cap: ceil (100)				
mean	10/10	4 [1, 6]	162 [42, 222]	2.87 [0.73, 3.95]
item-count-weighted mean	10/10	1 [1, 2]	24.9 [11.3, 37.7]	2.74 [1.24, 3.68]
geometric mean	10/10	4 [2, 6]	147 [38, 202]	3.21 [0.89, 4.11]
harmonic mean	10/10	4 [2, 6]	138 [38, 185]	3.71 [1.15, 4.34]
median	10/10	4.5 [2, 9]	4.73 [0.38, 16.64]	1.49 [0.38, 2.78]
min	10/10	2 [1, 3]	5.29 [2.22, 23.86]	3.83 [1.64, 12.07]
Borda	10/10	17.5 [10, 27]	54.6 [21.1, 100.5]	2.02 [1.21, 3.07]
Condorcet winner	10/10	17 [8, 22]	22.6 [4.6, 55.4]	1.74 [0.98, 3.03]
plurality	10/10	7 [4, 8]	5.18 [1.42, 11.42]	1.03 [0.50, 2.26]
cap: front+2				
mean	10/10	9.5 [2, 16]	162 [42, 222]	3.02 [0.77, 4.00]
item-count-weighted mean	10/10	7.5 [2, 13]	68.8 [14.7, 113.6]	2.83 [1.29, 4.08]
geometric mean	10/10	9 [3, 15]	150 [39, 206]	3.45 [0.93, 4.17]
harmonic mean	10/10	9 [3, 14]	143 [39, 194]	3.95 [1.21, 4.83]
median	10/10	4.5 [2, 9]	4.73 [0.38, 16.64]	1.49 [0.38, 2.78]
min	10/10	2 [1, 3]	5.29 [2.22, 23.86]	3.83 [1.64, 12.07]
Borda	10/10	17.5 [10, 27]	54.6 [21.1, 100.5]	2.02 [1.21, 3.07]
Condorcet winner	10/10	17 [8, 22]	22.6 [4.6, 55.4]	1.74 [0.98, 3.03]
plurality	10/10	7 [4, 8]	5.18 [1.42, 11.42]	1.03 [0.50, 2.26]
cap: gain+20				
mean	10/10	8.5 [3, 12]	162 [42, 222]	2.87 [0.73, 3.95]
item-count-weighted mean	10/10	2 [1, 4]	28.4 [11.3, 47.6]	2.74 [1.24, 3.68]
geometric mean	10/10	8 [3, 11]	148 [38, 202]	3.21 [0.89, 4.11]
harmonic mean	10/10	7.5 [3, 10]	139 [38, 185]	3.71 [1.15, 4.34]
median	10/10	4.5 [2, 9]	4.73 [0.38, 16.64]	1.49 [0.38, 2.78]
min	9/10	2 [1, 4]	3.01 [2.02, 23.58]	3.01 [1.21, 11.49]
Borda	10/10	18 [10, 28]	54.6 [21.1, 100.5]	2.02 [1.21, 3.07]
Condorcet winner	10/10	17 [8, 22]	22.6 [4.6, 55.4]	1.74 [0.98, 3.03]
plurality	10/10	7 [4, 8]	5.18 [1.42, 11.42]	1.03 [0.50, 2.26]
cap: gain+10				
mean	10/10	16.5 [5, 23]	162 [42, 222]	2.87 [0.73, 3.95]
item-count-weighted mean	10/10	5 [2, 9]	42.7 [12.2, 79.1]	2.74 [1.24, 3.68]
geometric mean	10/10	16 [4, 21]	155 [38, 209]	3.21 [0.89, 4.11]
harmonic mean	10/10	15.5 [4, 20.0125]	153 [38, 197]	3.71 [1.15, 4.34]
median	10/10	4.5 [2, 9]	4.73 [0.38, 16.64]	1.49 [0.38, 2.78]
min	7/10	2 [1, 3]	2.41 [2.02, 7.57]	2.41 [1.21, 4.64]
Borda	10/10	19.5 [10, 30]	54.6 [21.1, 100.5]	2.02 [1.21, 3.07]
Condorcet winner	10/10	17 [8, 22]	22.6 [4.6, 55.4]	1.74 [0.98, 3.03]
plurality	10/10	7 [4, 8]	5.18 [1.42, 11.42]	1.03 [0.50, 2.26]

Table 4: As the HELM top-10 table, for all 78 challengers and all six caps (appendix). Source: `results/helm_main.csv`; script `tab_rule_summary.py --targets all`.

rule	feasible	K	L_1	L_∞
cap: ceil (100)				
mean	78/78	16 [13, 19]	696 [562, 863]	13.2 [10.4, 16.4]
item-count-weighted mean	78/78	10 [7.5, 13]	281 [197, 408]	14.6 [10.9, 18.9]
geometric mean	78/78	16 [13, 19]	669 [535, 835]	14.2 [11.1, 17.5]
harmonic mean	78/78	16 [13, 20]	644 [511, 808]	15.3 [12.0, 18.7]
median	78/78	21 [18, 23.5]	112 [74, 163]	9.78 [7.74, 12.97]
min	78/78	11.5 [8.5, 15]	98.1 [57.1, 153.8]	22.2 [19.4, 24.5]
Borda	78/78	47 [44, 49]	534 [400, 696]	15.1 [11.4, 19.0]
Condorcet winner	78/78	29 [29, 29]	213 [161, 259]	10.2 [8.0, 13.1]
plurality	78/78	9 [9, 9]	47.2 [33.8, 59.6]	6.21 [5.27, 7.46]
cap: front+2				
mean	78/78	31 [27, 33.5]	696 [562, 863]	16.3 [12.1, 19.7]
item-count-weighted mean	78/78	27 [24, 30]	481 [365, 637]	17.5 [12.8, 22.9]
geometric mean	78/78	30.5 [26.5, 33]	677 [542, 843]	17.2 [12.9, 21.1]
harmonic mean	78/78	30 [26, 32.5125]	659 [526, 825]	18.9 [13.8, 22.3]
median	78/78	21 [18, 23.5]	112 [74, 163]	9.78 [7.74, 13.15]
min	78/78	11.5 [8.5, 15]	98.1 [57.1, 153.8]	22.2 [19.4, 24.5]
Borda	78/78	47 [44, 49]	534 [400, 696]	15.1 [11.4, 19.0]
Condorcet winner	78/78	29 [29, 29]	213 [161, 259]	10.2 [8.0, 13.1]
plurality	78/78	9 [9, 9]	47.2 [33.8, 59.6]	6.21 [5.27, 7.46]
cap: front+0				
mean	78/78	37 [33, 39.5]	696 [562, 863]	20.3 [15.7, 24.9]
geometric mean	78/78	36 [32, 38.5]	680 [546, 847]	21.5 [16.3, 25.5]
harmonic mean	78/78	35.5 [31, 38]	667 [533, 833]	22.4 [17.2, 26.7]
median	78/78	21 [18, 23.5]	116 [78, 191]	11.7 [9.7, 16.8]
min	78/78	11.5 [8.5, 15]	98.1 [57.1, 153.8]	22.2 [19.4, 24.5]
Borda	78/78	48 [46, 50]	549 [415, 714]	16.1 [12.3, 20.0]
Condorcet winner	78/78	31 [31, 32]	259 [198, 342]	11.9 [8.9, 15.2]
plurality	78/78	18 [18, 18]	149 [112, 180]	10.3 [8.1, 13.0]
cap: gain+20				
mean	56/78	28 [23.5, 33]	541 [453, 629]	9.92 [8.45, 11.77]
item-count-weighted mean	50/78	18 [11, 20]	290 [157, 322]	10.3 [7.0, 10.9]
geometric mean	50/78	25.5 [21, 29]	490 [405, 558]	10.4 [8.2, 11.1]
harmonic mean	50/78	25.5 [20, 29]	500 [390, 551]	11.2 [9.1, 12.0]
median	66/78	19 [16, 21.5]	80.1 [57.0, 119.1]	8.28 [6.82, 10.24]
min	34/78	6.5 [4, 8]	44.0 [23.4, 49.7]	11.7 [8.2, 15.6]
Borda	49/78	44 [40, 46]	331 [266, 403]	10.0 [7.4, 11.4]
Condorcet winner	65/78	29 [28, 29]	169 [134, 218]	8.30 [6.38, 10.45]
plurality	74/78	9 [9, 9]	45.5 [31.7, 52.9]	6.09 [4.98, 7.23]
cap: gain+10				
mean	29/78	37 [28, 46]	360 [276, 440]	6.46 [4.94, 8.16]
item-count-weighted mean	24/78	21 [9, 27]	190 [78, 250]	5.89 [3.61, 6.97]
geometric mean	24/78	32 [21, 43]	315 [209, 424]	5.97 [4.11, 8.01]
harmonic mean	20/78	28 [17, 38]	273 [168, 371]	5.78 [4.09, 7.58]
median	40/78	14 [11, 16]	43.0 [31.7, 60.8]	5.82 [4.76, 6.98]
min	14/78	3 [2, 5]	8.96 [2.41, 18.53]	5.14 [2.41, 7.64]
Borda	25/78	40 [30, 47]	185 [100, 280]	5.52 [3.06, 7.41]
Condorcet winner	38/78	28 [27, 28]	107 [86, 134]	5.83 [4.00, 6.78]
plurality	57/78	9 [9, 9]	31.0 [25.7, 40.4]	4.91 [3.58, 5.52]
cap: gain+5				
mean	11/78	34 [9, 53]	168 [42, 260]	2.98 [0.73, 4.62]
geometric mean	8/78	31.5 [6, 38]	154 [27, 187]	2.92 [0.60, 3.52]
harmonic mean	8/78	32 [6, 44]	157 [26, 218]	3.19 [0.73, 4.16]
median	16/78	9 [4, 10]	16.6 [3.3, 20.3]	2.89 [1.24, 3.82]
min	7/78	2 [1, 3]	2.41 [2.02, 7.57]	2.41 [1.21, 4.22]
Borda	11/78	24 [11, 45]	56.0 [21.1, 151.7]	2.04 [1.21, 4.00]
Condorcet winner	16/78	23.5 [16, 27]	56.5 [20.5, 73.5]	3.07 [1.63, 4.00]
plurality	29/78	9 [9, 10]	20.7 [15.2, 25.3]	2.99 [2.53, 3.54]

Table 5: As the main OLLB table, for all caps for which results exist (front+0, front+2, gain+5, gain+20, emp99; Borda, Condorcet winner and plurality only under ceil, front+2, gain+10 and emp95, and emp99 for the top 20 challengers). Source: `results/ollb_*.csv`; script `tab_rule_summary.py`.

rule	feasible	K	L_1	L_∞
cap: ceil (100)				
mean	10/10	1 [1, 1]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	10/10	1 [1, 1]	21.5 [3.7, 23.8]	6.04 [1.13, 6.97]
harmonic mean	10/10	1 [1, 1]	18.3 [2.0, 22.6]	7.04 [1.02, 8.88]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	10/10	2 [1, 2]	12.6 [3.7, 17.9]	8.05 [3.68, 10.81]
Borda	10/10	1 [1, 2]	5.67 [1.65, 8.80]	2.31 [0.68, 4.92]
Condorcet winner	10/10	2 [2, 3]	10.6 [6.7, 13.7]	5.31 [3.27, 6.20]
plurality	10/10	2 [2, 2]	15.4 [7.5, 20.1]	11.4 [4.1, 12.3]
cap: front+2				
mean	10/10	1 [1, 2]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	10/10	1 [1, 2]	21.5 [3.7, 23.8]	6.04 [1.13, 7.07]
harmonic mean	10/10	1.5 [1, 2]	18.3 [2.0, 22.6]	7.04 [1.02, 8.97]
trimmed mean	10/10	–	21.3 [7.4, 23.0]	5.32 [1.85, 5.76]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	10/10	2 [1, 2]	12.6 [3.7, 17.9]	8.05 [3.68, 10.81]
Borda	10/10	1 [1, 2]	5.67 [1.65, 8.80]	2.31 [0.68, 4.92]
Condorcet winner	10/10	2 [2, 3]	10.6 [6.7, 13.7]	5.31 [3.27, 6.20]
plurality	10/10	2 [2, 2]	15.4 [7.5, 20.1]	11.4 [4.1, 12.3]
cap: front+0				
mean	10/10	1 [1, 2]	25.0 [7.9, 28.3]	4.30 [1.32, 4.71]
geometric mean	10/10	1.5 [1, 2]	21.5 [3.7, 23.8]	6.04 [1.13, 7.23]
harmonic mean	10/10	1.5 [1, 2]	18.3 [2.0, 22.6]	7.04 [1.02, 9.03]
trimmed mean	10/10	–	21.3 [7.4, 23.0]	5.32 [1.85, 5.76]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	10/10	2 [1, 2]	12.6 [3.7, 17.9]	8.05 [3.68, 10.81]
cap: gain+20				
mean	10/10	2 [1, 2]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	10/10	2 [1, 2]	21.5 [3.7, 23.8]	6.04 [1.13, 6.97]
harmonic mean	10/10	2 [1, 2]	18.3 [2.0, 22.6]	7.04 [1.02, 8.88]
trimmed mean	10/10	–	21.3 [7.4, 23.0]	5.32 [1.85, 5.76]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	10/10	2 [1, 2]	12.6 [3.7, 17.9]	8.05 [3.68, 10.81]
cap: gain+10				
mean	10/10	3 [1, 3]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	10/10	3 [1, 3]	23.5 [3.7, 28.8]	6.04 [1.13, 6.97]
harmonic mean	10/10	2 [1, 4]	18.3 [2.0, 34.5]	7.04 [1.02, 8.88]
trimmed mean	10/10	–	21.3 [7.4, 23.0]	5.32 [1.85, 5.76]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	7/10	2 [1, 2]	6.25 [3.12, 13.15]	5.25 [3.12, 8.16]
Borda	10/10	1.5 [1, 2]	5.67 [1.65, 8.80]	2.31 [0.68, 4.92]
Condorcet winner	10/10	3 [2.5, 3]	10.6 [6.7, 13.7]	5.31 [3.27, 6.20]
plurality	3/10	2 [2, 2]	5.55 [3.50, 7.47]	3.23 [2.19, 4.12]
cap: gain+5				
mean	10/10	5.5 [2, 6]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	4/10	1 [1, 6]	3.30 [2.44, 28.36]	1.05 [0.79, 4.85]
harmonic mean	4/10	1 [1, 6]	1.84 [1.28, 28.77]	0.94 [0.67, 4.95]
trimmed mean	3/10	–	5.78 [4.60, 7.38]	1.44 [1.15, 1.85]
median	10/10	1 [1, 2]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	3/10	1 [1, 1]	3.12 [2.79, 3.68]	3.12 [2.79, 3.68]

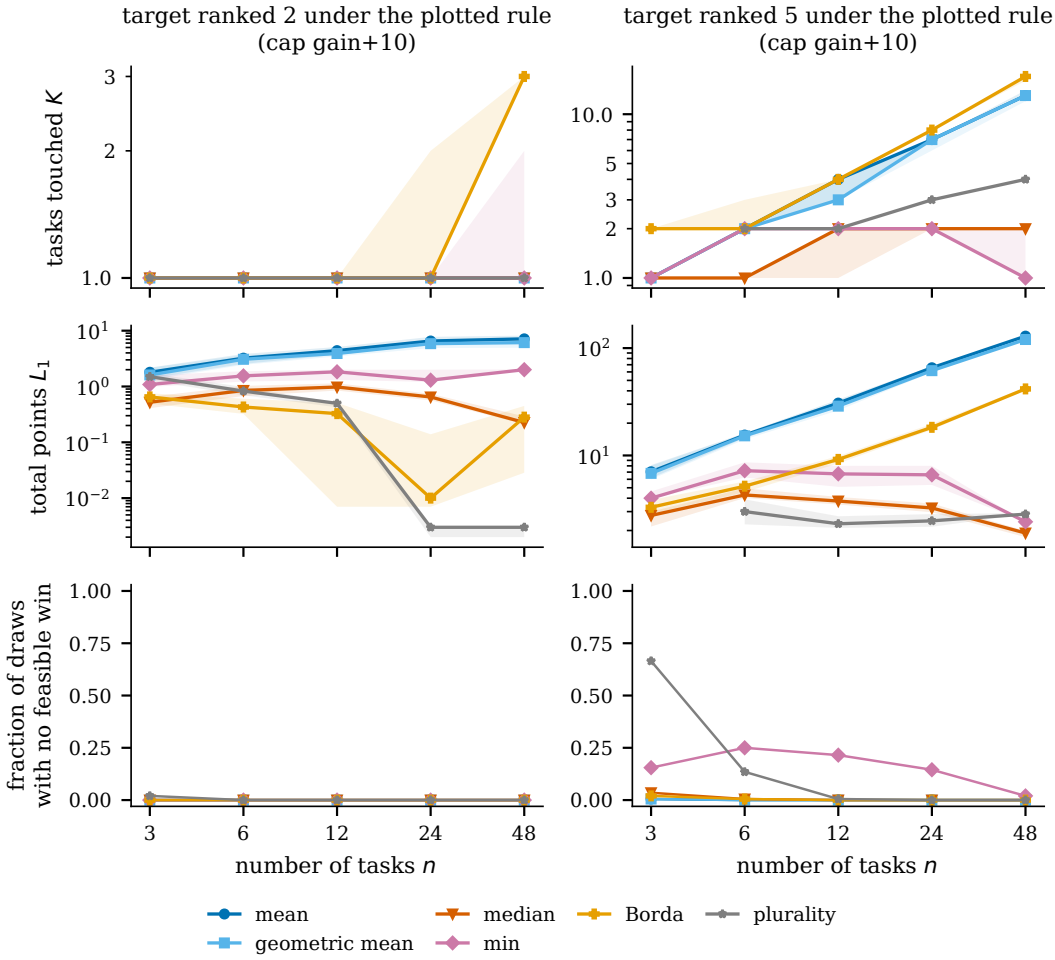


Figure 9: Investment against the number of tasks n, HELM MMLU, cap *gain+10*: n subjects are drawn at random from the 57 (200 draws per n). In every draw and for every rule the targets are the models ranked 2 and 5 *under that rule*, so different rules are evaluated on different models within a draw and the rules are not compared on like-for-like targets. Lines: median over draws, with 95% percentile bootstrap bands over draws. Top: tasks touched K; middle: total points L_1 (draws in which the target already wins cost 0 and cannot be shown on the log axis); bottom: fraction of draws in which first place is infeasible under the cap (medians are drawn only where finite). Source: `results/summary_helm_tasksub_gain10_by_n.csv`; script `fig_taskscale.py`.

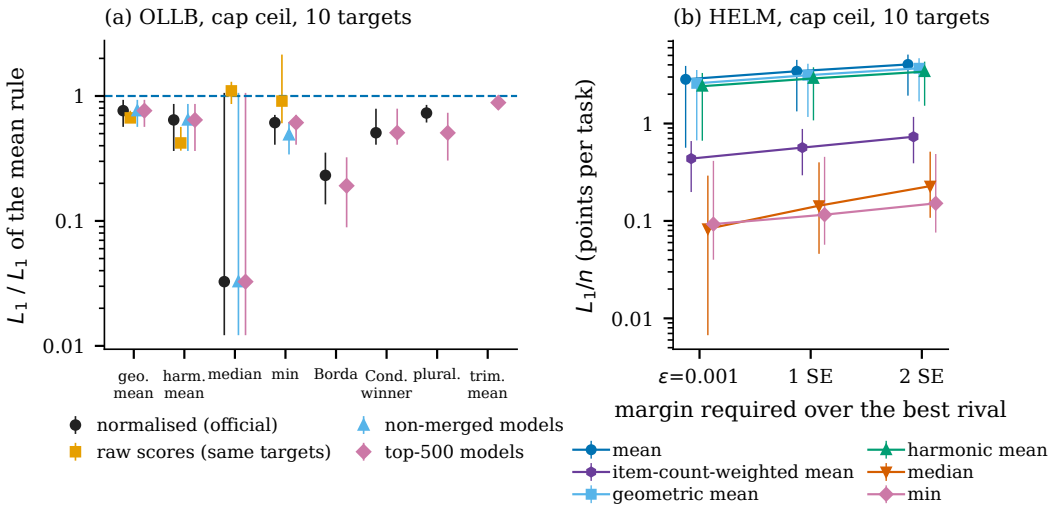


Figure 10: (a) Open LLM Leaderboard, cap ceil, ten challengers ranked 2–11 under the mean rule of each variant: per-target ratio $L_1^{\text{rule}}/L_1^{\text{mean}}$ (median, 95% bootstrap over the 10 targets; dashed: mean rule) for the official normalised scores, the raw scores (on the same ten targets as the normalised ones), the non-merged models and the top-500 models. Rules without a marker were not run on that variant. Rank rules depend on ranks only, so they are unaffected by raw vs normalised scoring. (b) HELM, cap ceil, ten challengers: median L_1/n (points per task) with 95% bootstrap over targets when first place requires a margin over the best rival of $\varepsilon = 0.001$, 1 SE or 2 SE (SE = 0.6 points, the median standard error of a difference of two models' equal-weight means from the binomial sampling of the per-subject item counts); the margin is only an approximate scale for median and min; no margin variant exists for the rank rules. Source: `results/summary_ollb{,_nomerge,_top500}_ratio_to_mean.csv`, `summary_ollb_raw_vs_norm_paired.csv`, `summary_helm_{main,margin1se,margin2se}_by_rule_group` script `fig_variants.py`.

Table 6: Minimum investment for first place on the Open LLM Leaderboard (6 tasks, 4496 models), ten challengers ranked 2–11 under the mean rule, caps ceil, emp95 (per-task 95th percentile of observed fine-tune gains) and gain+10: median [95% bootstrap CI over targets] of tasks touched K, total points L_1 and largest single-task jump L_∞ over targets for which first place is attainable; "feasible" is the number of attainable targets out of 10 (infeasible targets are counted, not imputed; medians of cells with different feasible counts are not comparable). Borda, Condorcet winner and plurality are run on sampled targets that include these ten (no emp99 shown here). K was not recorded for the trimmed mean. Other caps are in the appendix table. Source: `results/ollb_closed.csv`, `ollb_score_caps.csv`, `ollb_score_emp.csv`, `ollb_ordinal_*.csv`; script `tab_rule_summary.py`.

rule	feasible	K	L_1	L_∞
cap: ceil (100)				
mean	10/10	1 [1, 1]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	10/10	1 [1, 1]	21.5 [3.7, 23.8]	6.04 [1.13, 6.97]
harmonic mean	10/10	1 [1, 1]	18.3 [2.0, 22.6]	7.04 [1.02, 8.88]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	10/10	2 [1, 2]	12.6 [3.7, 17.9]	8.05 [3.68, 10.81]
Borda	10/10	1 [1, 2]	5.67 [1.65, 8.80]	2.31 [0.68, 4.92]
Condorcet winner	10/10	2 [2, 3]	10.6 [6.7, 13.7]	5.31 [3.27, 6.20]
plurality	10/10	2 [2, 2]	15.4 [7.5, 20.1]	11.4 [4.1, 12.3]
cap: emp95				
mean	10/10	2 [1, 2]	25.0 [7.9, 28.3]	4.49 [1.32, 5.32]
geometric mean	9/10	4 [1, 5]	38.2 [2.9, 49.3]	8.93 [0.96, 15.54]
harmonic mean	4/10	1 [1, 3]	1.84 [1.28, 19.83]	0.94 [0.67, 5.78]
trimmed mean	10/10	–	21.3 [7.4, 23.0]	6.64 [1.85, 7.69]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	4/10	1 [1, 2]	3.40 [2.79, 6.25]	3.40 [2.79, 5.25]
Borda	10/10	2 [1, 2]	5.68 [1.65, 8.89]	2.31 [0.68, 5.09]
Condorcet winner	5/10	3 [2, 3]	6.70 [5.29, 11.58]	3.90 [2.37, 5.36]
plurality	9/10	2 [2, 2]	15.4 [10.7, 21.4]	11.3 [8.3, 12.3]
cap: gain+10				
mean	10/10	3 [1, 3]	25.0 [7.9, 28.3]	4.16 [1.32, 4.71]
geometric mean	10/10	3 [1, 3]	23.5 [3.7, 28.8]	6.04 [1.13, 6.97]
harmonic mean	10/10	2 [1, 4]	18.3 [2.0, 34.5]	7.04 [1.02, 8.88]
trimmed mean	10/10	–	21.3 [7.4, 23.0]	5.32 [1.85, 5.76]
median	10/10	1 [1, 1]	0.89 [0.35, 8.37]	0.45 [0.17, 4.18]
min	7/10	2 [1, 2]	6.25 [3.12, 13.15]	5.25 [3.12, 8.16]
Borda	10/10	1.5 [1, 2]	5.67 [1.65, 8.80]	2.31 [0.68, 4.92]
Condorcet winner	10/10	3 [2.5, 3]	10.6 [6.7, 13.7]	5.31 [3.27, 6.20]
plurality	3/10	2 [2, 2]	5.55 [3.50, 7.47]	3.23 [2.19, 4.12]

Table 7: Open LLM Leaderboard (6 tasks, 4496 models), ten challengers ranked 2–11 under the mean rule. (a) Median [95% bootstrap CI over the 10 targets] of the per-target ratio $L_1^{\text{rule}}/L_1^{\text{mean}}$ at cap ceil (a ratio < 1 means cheaper than the mean rule), under the official normalised scores, under the raw scores for the same ten targets (ranked 2–11 on normalised scores), and the paired per-target difference raw $-$ normalised (median [95% CI]; an interval containing 0 means the sign is not established with 10 targets); then after removing merged models and restricted to the 500 best models (each ranked within its own slice). (b) Number of the ten targets for which first place is attainable inside the cap (infeasible targets are counted here, not dropped; the medians in (a) are over targets feasible under both rules, all of them at ceil). gain+h: a score rises by at most h points; emp95/emp99: per task, the 95th/99th percentile of the 540 observed fine-tune minus base gains. "–": not run (rank-based rules do not depend on the scoring scale, and Borda, Condorcet and plurality were run on a sample of 102 targets that contains these ten, only under ceil, front+2, gain+10, emp95 (and emp99 for the top 20); the trimmed mean has no ceil run). Identical entries in the normalised, non-merged and top-500 columns mean the ten targets and their costs coincide. The median rule's advantage under normalised scores (0.03) is largely absent under raw scores (1.10); for these ten targets the paired difference is large (+0.93) but its interval includes 0 (7 of 10 targets dearer under raw), whereas for ranks 11–100 it is established in the paired analysis of all targets in the source file. Source: `results/summary_ollb{,_nomerge,_top500}_ratio_to_mean.csv`, `summary_ollb_raw_vs_norm_paired.csv`, `summary_ollb_by_rule_group.csv`; script `tab_ollb_robust.py`.

	(a) L_1/L_1^{mean} at cap ceil, median [95% CI]					(b) targets feasible at cap			
rule	normalised	raw	raw − norm.	non-merged	top-500	gain+5	gain+10	emp95	emp99
mean	1 (ref.)	1 (ref.)	–	1 (ref.)	1 (ref.)	10/10	10/10	10/10	10/10
geometric mean	0.76 [0.57, 0.93]	0.67 [0.62, 0.76]	-0.13 [-0.19, +0.14]	0.76 [0.57, 0.93]	0.76 [0.57, 0.93]	4/10	10/10	9/10	10/10
harmonic mean	0.64 [0.36, 0.86]	0.42 [0.37, 0.57]	-0.27 [-0.33, +0.13]	0.64 [0.36, 0.86]	0.64 [0.36, 0.86]	4/10	10/10	4/10	10/10
trimmed mean	–	–	–	–	0.88 [0.78, 0.96]	3/10	10/10	10/10	10/10
median	0.03 [0.01, 1.06]	1.10 [0.86, 1.30]	+0.93 [-0.15, +1.27]	0.03 [0.01, 1.06]	0.03 [0.01, 1.06]	10/10	10/10	10/10	10/10
min	0.61 [0.41, 0.71]	0.91 [0.61, 2.15]	+0.25 [+0.14, +1.56]	0.49 [0.34, 0.62]	0.61 [0.41, 0.71]	3/10	7/10	4/10	5/10
Borda	0.23 [0.14, 0.35]	–	–	–	0.19 [0.09, 0.32]	–	10/10	10/10	–
Condorcet winner	0.51 [0.41, 0.79]	–	–	–	0.51 [0.41, 0.79]	–	10/10	5/10	–
plurality	0.73 [0.61, 0.85]	–	–	–	0.51 [0.30, 0.73]	–	3/10	9/10	–

Table 8: Rank-based rules on the Open LLM Leaderboard (6 tasks): Borda, Condorcet winner (existence of a Condorcet winner, pairwise majority) and plurality, minimum investment for first place by cap. Targets are a sample of 102 models (the 20 top challengers and 100 log-rank-uniform ones, with overlap), so the infeasible counts describe that sample only, not the 4495 challengers; emp99 was run for the top 20 only. "infeasible": no investment inside the cap gives first place (infinite cost); "time limit": solver time limit (one Condorcet target at ceil, one at front+2, where only K is missing), counted separately and not as infeasible; the medians of K and L_1 are over the finite targets ("finite") and therefore not comparable across caps with different counts. Source: `results/ollb_ordinal_{ceil,front2,gain10,emp95,emp99_top20}.csv`; script `tab_ollb_ordinal.py`.

rule	targets	infeasible	time limit	median K	median L_1	finite
cap: ceil (100)						
Borda	102	0	0	3.00	23.3	102
Condorcet winner	102	0	1	4.00	36.0	101
plurality	102	0	0	2.00	26.4	102
cap: front+2						
Borda	102	0	0	3.00	23.3	102
Condorcet winner	102	0	1	4.00	33.2	101
plurality	102	0	0	2.00	26.4	102
cap: gain+10						
Borda	102	37	0	2.00	12.4	65
Condorcet winner	102	62	0	4.00	18.3	40
plurality	102	89	0	2.00	10.4	13
cap: emp95						
Borda	102	43	0	2.00	11.9	59
Condorcet winner	102	86	0	3.00	11.3	16
plurality	102	63	0	2.00	22.8	39
cap: emp99						
Borda	20	0	0	1.00	5.99	20
Condorcet winner	20	1	0	3.00	11.1	19
plurality	20	2	0	2.00	16.1	18